

UI/UX Tactical Design System

Project: Project Rudra-Logistics

Interface Style: Military-Grade Tactical Command & Control (C2) HUD

Form Factor: Responsive Desktop / Command Room Large Displays (1920x1080 minimum target)

1. Visual Design Philosophy & Aesthetics

The design of Project Rudra-Logistics avoids commercial e-commerce aesthetics. Instead, it mirrors modern military situational awareness systems (such as AFMS, C4ISR, and NATO Blue Force Trackers):

1. **Dark Matte Surfaces:** Deep slate and charcoal surfaces minimize glare and optical fatigue in low-light tactical command bunkers.
 2. **High-Contrast Telemetry:** Crucial status indicators pop vividly against dark backgrounds with clear semantic coloring (Emerald, Amber, Crimson, Cyan).
 3. **Information Density with Zero Clutter:** Every pixel conveys actionable operational data: coordinates, temperatures, Days of Supply, and convoy movement vectors.
 4. **Monospaced Data Alignment:** All quantitative telemetry is rendered in monospaced fonts to ensure rapid, jitter-free reading by logistics staff.
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2. Color Palette & Semantic Tokens

Token Name	Hex Code	Tailwind Equivalent	Purpose / Usage
Base Surface	#090d16	bg-slate-950	Primary application canvas background
Panel Surface	#0f172a	bg-slate-900	Card, sidebar, and floating HUD modal background
Border & Divider	#1e293b	border-slate-800	Subtle boundaries between modular panels
HUD Accent (Cyan)	#06b6d4	text-cyan-400	Active filters, selected routes, GPS coordinates
Tactical Olive	#4d7c0f	text-lime-700	Military branding accents and secondary buttons
Status Nominal (Green)	#10b981	text-emerald-400	Healthy stock ($\text{DOS} > 15\%$), open pass, normal temp
Status Warning (Amber)	#f59e0b	text-amber-400	Medium stock ($5\% < \text{DOS} \leq 15\%$), pass caution alert
Status Critical (Red)	#f43f5e	text-rose-500	Critical shortage ($\text{DOS} \leq 5\%$), blocked pass, cold-chain breach

3. Typography & Hierarchy

- **Primary Interface Font:** Inter, -apple-system, sans-serif (Clean, legible labels).
- **Telemetry & Numbers:** JetBrains Mono, Roboto Mono, monospace (Grid-aligned numerals).
- **Text Scale:**
- **Screen / Sector Title:** text-lg font-bold tracking-wider uppercase
- **Metric Stat Value:** text-2xl font-mono font-bold tracking-tight
- **Label / Caption:** text-xs font-mono uppercase tracking-widest text-slate-400
- **Military Timestamps:** text-xs font-mono text-cyan-400

4. Master Screen Layout Structure

The layout is arranged into a dense 4-quadrant Command Operating Picture (COP):

1. TACTICAL COMMAND HEADER [INDIAN ARMY LOGISTICS] [SECTOR: 14 CORPS LEH] [ZULU: 09:41:20Z] [DEFCON: 2]	
2. INTERACTIVE 3D GIS CORRIDOR MAP - Leh Base Depot (Node) - Khardung La Pass (● / ● Toggle) - Partapur Transit Camp - Siachen Base Camp (FOP) - Moving Convoy Blips (Live GPS)	3. DOS & OUTPOST INVENTORY GRID [SIACHEN BASE CAMP] - Kerosene Heating: ● 3.2 Days - High Alt Rations: ● 21.0 Days - 5.56mm Ammo: ● 8.5 Days [DAULAT BEG OLDI] - Kerosene Heating: ● 19.5 Days
4. CONVOY TELEMETRY & COLD-CHAIN HUD Convoy: STALLION-ALPHA (14C/TRK-09) - Location: 34.279 N, 77.604 E - Cargo: Plasma (+4.2°C ●) - Fuel Tank: 68% Speed: 32 km/h	5. "WHAT-IF" WAR-GAMING SANDBOX [SLIDER] Temp: -30°C [Spike: +240%] [SLIDER] Mobilization: +50% Troops [TOGGLE] Block Khardung La Pass [ACTION] RECALCULATE LOGISTICS CHAIN

5. Key Component Specifications

5.1 Tactical Command Header

- **Military Clock:** Displays both Local IST and Military Zulu (UTC) time with a pulsing synchronization indicator.
- **Threat Indicator Badge:** Toggleable badge (PEACE / HEIGHTENED / ACTIVE ENGAGEMENT) that changes system-wide demand multipliers.
- **Critical Alerts Ticker:** Scrolling marquee with red flasher for real-time roadblock notifications (e.g., "ALERT: AVALANCHE AT Khardung La - CONVOY ALPHA RE-ROUTING VIA SHYOK").

5.2 3D GIS Map Viewport

- **Layer Toggles:**
 - Terrain Elevation & Slope.
 - Active Convoy Vectors with dotted line trajectory.
 - Mountain Passes with Weather Hazard Icons (Snowflake, Landslide, Wind).
- **Interactive Roadblock Switch:** Clicking on any mountain pass opens a contextual prompt: [SET STATUS: BLOCKED / CLEAR] . Toggling automatically triggers path re-calculation.

5.3 Days of Supplies (DOS) Matrix Card

- Each forward post card displays:
- Post Name & Altitude badge (12,000 FT).
- Current ambient temperature with dynamic thermometer icon.
- Mini progress bars for the 4 supply classes.
- Quick Action button: [REQUEST AIRDROP] or [DISPATCH CONVOY] .

5.4 IoT Cold-Chain Gauge

- Renders an analog-style dial or digital gauge for Class VIII medical containers.
- Visual Thresholds:
- $< 2^{\circ}\text{C}$: Blue warning (Risk of freezing).
- $2^{\circ}\text{C} - 8^{\circ}\text{C}$: Solid green (Safe medical window).
- $> 8^{\circ}\text{C}$: Red flashing alarm (Spoilage danger).

5.5 "What-If" War-Gaming Modal

- Allows tactical planners to test hypotheses before winter passes close.
- Visual outputs: Depletion forecast graph comparing *Baseline Supply* vs *War-Game Stress Scenario*.

5.6 Master Admin Operational Console (The Control Plane)

- Accessible via a dedicated top-bar navigation toggle ([⚙ MASTER ADMIN C2]) protected by Admin credential authorization.
- **Controls & Levers:**
- **Global & Sector Multiplier Sliders:** Real-time adjustments to standard Bukhari kerosene burn rates (L/soldier/day), vehicle fuel efficiency loss curves, and ration buffers.
- **Manual Pass State Overrides:** Instant force-toggle for mountain passes (Khardung La, Chang La, Rohtang) with one-click broadcast to all field logistics officers.
- **Emergency Inventory Injection Modal:** Allows the administrator to log sudden air-dropped relief packages (tonnes of food/fuel) into forward posts.
- **User Clearance & Credential Management:** Toggle active statuses, assign tactical roles (Commander, Logistics Officer, Auditor), and revoke credentials.

5.7 Cyber Defense & Anti-Spoofing HUD

- **Cryptographic Audit Badge:** Prominent green shield indicator (SHIELD: VERIFIED (0 TAMPER BLOCKS)) at top right. Clicking it opens the **Audit Chain Explorer** showing transaction block hashes, timestamped officer signatures, and unbroken SHA-256 links.

- **GPS Spoofing Threat Banner:** When an abnormal telemetry packet (impossible mountain speeds or coordinate teleportation) is received, the convoy marker on the map blinks in fluorescent violet with a pulsing warning:  GPS SPOOF DETECTED – FALLING BACK TO DEAD RECKONING .
- **System Hardening Status Card:** Live metrics displaying active JWT sessions, failed login attempts (blocked by sliding window rate limiter), and air-gapped network isolation confirmation.